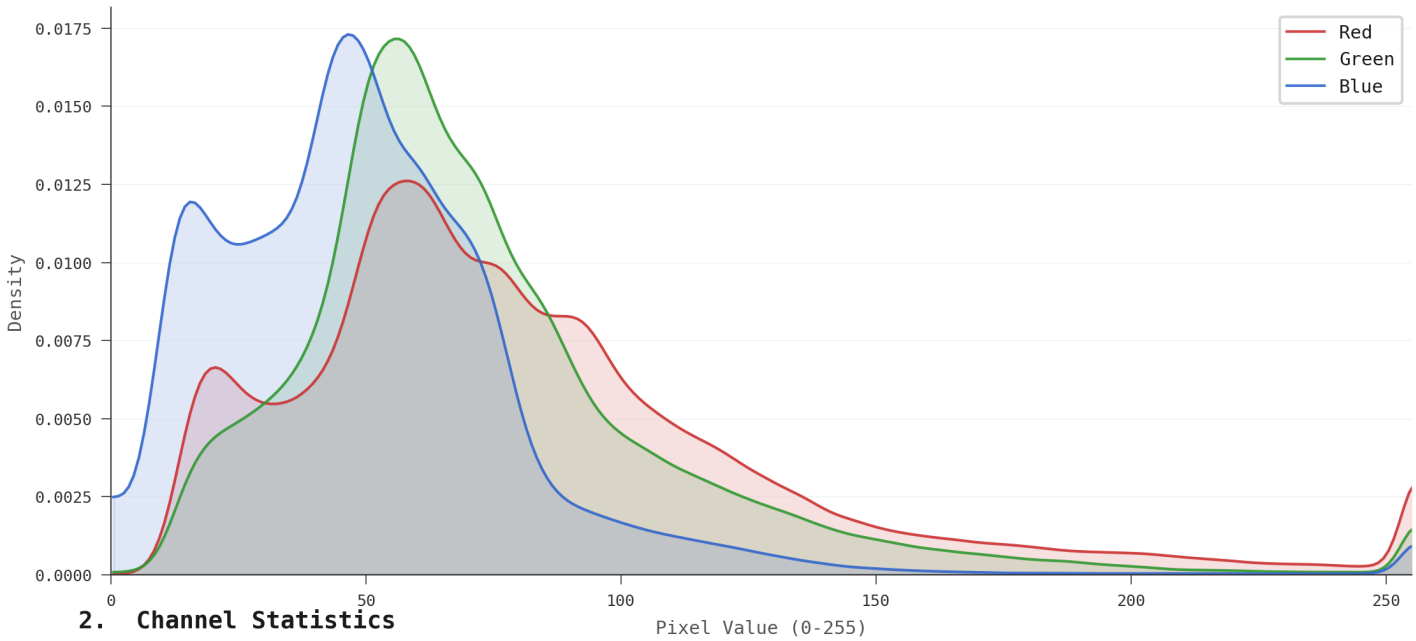
Reference Data Sheet: KODIM18

Subject: Fashion Portrait Woman

512x768 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	80.5698	47.3467	2241.7069	2.1586	1.3143	0	255
Green	71.8574	37.8520	1432.7776	4.2459	1.6283	0	255
Blue	49.5295	30.7754	947.1238	9.5076	2.0417	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.9158	0.7526
G	0.9158	1.0000	0.8352
B	0.7526	0.8352	1.0000

Pairwise Pearson Correlation Coefficients

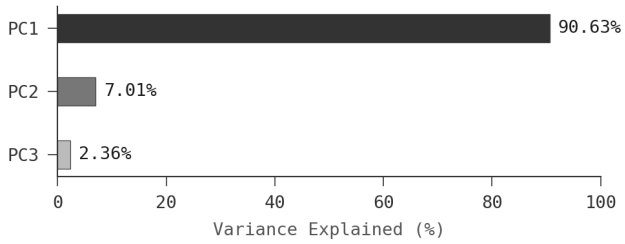
r(R-G)	=	**0.915823**
r(R-B)	=	**0.752554**
r(G-B)	=	**0.835186**
Avg |r|	=	**0.834521**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{2241.7126} & 1641.3123 & 1096.5581 \\
1641.3123 & \mathbf{1432.7812} & 972.9193 \\
1096.5581 & 972.9193 & \mathbf{947.1262}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM18

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	**-0.7117**	-0.5692	-0.4116	4188.5703
PC2	-0.5547	0.0959	0.8265	324.0742
PC3	0.4310	-0.8166	0.3840	108.9754

6. Derived Metrics

Condition Number

38.44

(l1/l3) PC1/PC2: 12.92 | PC2/PC3: 2.97

Blue Channel Independence

25.1%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Red

Loading: -0.7117 | Red dominant

Classification: Weakly One-Dimensional (PC1 = 90.63%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.925773	0.935640	0.930707
Green	0.879187	0.894688	0.886937
Blue	0.829050	0.848442	0.838746

8. Average Local Variance (3x3)

Channel	Value
Red	232.2003
Green	238.0022
Blue	212.5200

9. Redundancy Profile

Classification: Weakly One-Dimensional | Condition Number: 38.44 | PC1: 90.63%

Correlation: R–G pair leads at r = 0.915823 | G–B: 0.835186 | R–B: 0.752554

Blue Channel: 25.1% independent | Substantial independence from PC1

Spatial: Avg autocorrelation: 0.885463 | Avg local variance: 227.5742

Eigenvector: Red dominant | R (-0.7117) leads | G (-0.5692), B (-0.4116)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."